ProjFlow: Projection Sampling with Flow Matching for Zero-Shot Exact Spatial Motion Control

Anonymous CVPR submission

Abstract

We present ProjFlow, a training-free sampling method for imposing spatial control on text-conditioned human motion generation with a pretrained flow-matching prior. The key observation is that many motion control tasks can be written as linear measurement models, which allows each sampling step to project the model's clean motion estimate onto the feasible set in closed form. ProjFlow performs this correction in a kinematics-aware metric induced by a skeleton Laplacian so that updates spread coherently across connected joints, then recomposes the corrected endpoint with noise to remain on the motion manifold. For sparse controls such as long-gap inpainting, we further introduce time-varying pseudo-observations with dynamic masking and adaptive uncertainty. On HumanML3D, ProjFlow achieves exact trajectory satisfaction with zero-shot sampling, reaching 0.0000 *trajectory, location, and average errors for both pelvis-only and all-joint control, while maintaining competitive realism with FID* 0.107 *and* 0.097*, respectively. On 2D-to-3D lifting, it achieves exact reprojection error (*0.000 *MPJPE-2D) and improves over Sketch2Anim in FID (*$0.349/0.168$ *vs.* $0.525/0.577$ *under the Average/Cross protocols). Ablations show that the kinematics-aware metric, stochastic recomposition, and pseudo-observations are all important for preserving realism even though hard feasibility is maintained throughout.*

1. Introduction

Text-conditioned human motion generation has improved rapidly with diffusion and flow-based priors, making it possible to synthesize realistic motions from open-vocabulary prompts [2, 4, 9, 11, 16, 20]. The remaining bottleneck is control. In animation and interaction design, users often need motions that follow prescribed joint trajectories, satisfy keyframes, or agree exactly with external observations such as 2D sketches. Existing solutions typically fall into two groups. The first trains dedicated controllers or adapters for a particular control interface [3, 12, 18, 21]. These methods can be effective, but each new control type usually requires paired data and retraining. The second keeps the prior fixed and performs inference-time optimization [6, 13]. This is more flexible, but optimization is slower and generally does not guarantee exact feasibility at every step.

This paper studies a narrower but practically important regime: spatial control problems that admit a linear observation model. That assumption covers dense and sparse trajectory control, motion inpainting with masked coordinates, and 2D-to-3D lifting under a linear camera model. Once written as $y = Ax + \epsilon$, these tasks can be handled by a closed-form projection rather than iterative latent optimization. The challenge is to do so without damaging the motion prior. A naive Euclidean projection enforces the measurements but tends to create implausible corrections, especially when a sparse constraint is applied to one joint and the rest of the body must respond naturally.

ProjFlow addresses this by integrating a weighted projection directly into each flow-matching sampling step. The sampler first recovers the model's clean endpoint estimate, then projects that endpoint using an R-norm whose geometry is defined by the kinematic tree. Corrections therefore propagate along the skeleton rather than affecting joints independently. The projected endpoint is then recomposed with a noise estimate in the spirit of posterior-sampling methods for inverse problems, which keeps the trajectory close to the learned motion manifold. For sparse and long-range controls, ProjFlow augments the hard observations with time-varying pseudo-observations and adaptive uncertainty, providing guidance early in sampling while preserving exact hard constraints.

Our contributions are threefold. First, we cast a range of spatial motion control tasks as linear inverse problems for pretrained flow-matching motion priors and derive a closed-form projection sampler that requires no retraining. Second, we introduce a kinematics-aware metric and a time-varying pseudo-observation scheme that substantially improve realism under exact constraints. Third, we show on HumanML3D that zero-shot projection sampling can match or exceed the realism of several specialized controllers while uniquely achieving exact feasibility across trajectory control and 2D-to-3D lifting.

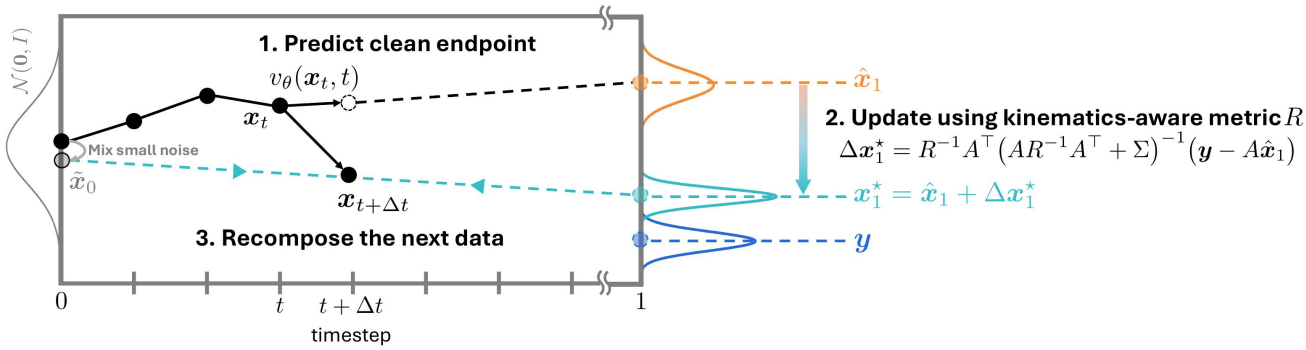


Figure 1. **Overview of the Projection Sampling Step.** At each flow step, ProjFlow predicts a clean endpoint from the current noisy motion, projects that endpoint onto the measurement-consistent set with a closed-form R-metric update, and recomposes the next noisy state with stochastic mixing. For sparse controls, pseudo-observations enlarge the support of the constraints early in sampling and fade toward the exact hard observations.

2. Related Work

Human motion priors. Large-scale text-to-motion benchmarks such as HumanML3D enabled strong conditional motion generators [4]. Diffusion-based models including MDM, MotionDiffuse, and MoFusion established high-quality text-conditioned motion synthesis [2, 16, 20]. More recent work argues that absolute world-space coordinates simplify downstream control and improve fidelity for motion generation [11]. ProjFlow builds on this setting: it does not change the pretrained prior, but instead modifies the sampler.

Controllable motion generation. Prior work on controllable motion typically trains a model for a particular conditioning mechanism. GMD introduces guided diffusion for spatially constrained motion synthesis [5]. OmniControl extends this idea to arbitrary joints and times through analytic spatial guidance and realism guidance [18]. MotionLCM combines latent consistency distillation with a ControlNet-style controller for fast inference [3, 19], while ControlMM studies masked motion generation for flexible control signals [12]. Sketch2Anim learns a sketch-conditioned pipeline for storyboard-to-motion generation [21]. These approaches are effective but task- or interface-specific. By contrast, ProjFlow is a training-free sampler that handles any control signal that can be expressed linearly.

Inference-time optimization and inverse problems. A different line of work uses a pretrained motion prior and optimizes during inference. PriorMDM composes pretrained diffusion priors for editing and control [13], while DNO directly optimizes diffusion noise to satisfy user-defined objectives [6]. Outside motion, diffusion inverse-problem solvers such as DDNM, PIGDM, DPS, DDRM, and ReSample show that pretrained generative models can enforce measurement consistency without task-specific retraining [1, 7, 14, 15, 17]. FlowDPS extends this perspective to flow models [8]. ProjFlow is most closely related in spirit to this literature, but adapts the idea to articulated motion and introduces a kinematics-aware geometry that is specific to skeletal constraints.

3. Method

3.1. Problem Setup

We represent a motion sample as $x \in \mathbb{R}^{3\times L\times J}$, where L is the number of frames and J is the number of joints. ProjFlow assumes a pretrained text-conditioned flow-matching model that predicts a velocity field $v_\theta(x_t, t, c)$ for prompt c [9, 10]. Following the implementation, the sampler recovers the clean endpoint estimate $\hat{x}_1$ and the noise endpoint estimate $\hat{x}_0$ from the current state x_t and the path coefficients α_t, σ_t:

$$\hat{x}_1 = \frac{\dot{\sigma}_t x_t - \sigma_t v_\theta(x_t, t, c)}{\alpha_t\dot{\sigma}_t - \sigma_t\dot{\alpha}_t}, \qquad \hat{x}_0 = \frac{-\dot{\alpha}_t x_t + \alpha_t v_\theta(x_t, t, c)}{\alpha_t\dot{\sigma}_t - \sigma_t\dot{\alpha}_t}. \tag{1}$$

For the linear rectified-flow path, this reduces to $\hat{x}_1 = x_t + (1-t)v_\theta$ and $\hat{x}_0 = x_t - tv_\theta$.

We consider control tasks that can be written as a linear observation model

$$y = Ax + \epsilon, \tag{2}$$

where A is a known measurement operator. Examples include selecting constrained coordinates for trajectory control, masking observed entries for inpainting, and orthographic 3D-to-2D projection for lifting. The goal is to sample motions that remain realistic under the pretrained prior while satisfying the measurements exactly when the observation noise vanishes.

3.2. Projection Sampling in Clean-Endpoint Space

Rather than modifying the latent noise with iterative optimization, ProjFlow corrects the clean endpoint estimate directly. Given $\hat{x}_1$, we solve the weighted projection

$$x_1^\star = \arg\min_x \frac{1}{2}\|x - \hat{x}_1\|_R^2 \quad \text{s.t.} \quad Ax = y \tag{3}$$

with $\|u\|_R^2 = u^\top R u$. The resulting closed-form update is

$$x_1^\star = \hat{x}_1 + R^{-1}A^\top(AR^{-1}A^\top)^{-1}(y - A\hat{x}_1). \tag{4}$$

To support soft pseudo-observations or noisy measurements, we use the more general form

$$x_1^\star = \hat{x}_1 + R^{-1}A^\top(AR^{-1}A^\top + \Sigma)^{-1}(y - A\hat{x}_1), \tag{5}$$

where Σ is diagonal and becomes zero on hard constraints. Eq. (5) is the core ProjFlow update shown in Fig. 1. When $\Sigma \to 0$, the selected measurements are satisfied exactly.

3.3. Kinematics-Aware Metric

A Euclidean projection treats each coordinate independently and often produces implausible motion edits. We instead define the metric through the kinematic tree. Let $L_{\text{kin}} \in \mathbb{R}^{J\times J}$ be the combinatorial Laplacian of the skeleton graph. ProjFlow uses

$$R = I_3 \otimes I_L \otimes (w_{\text{kin}} L_{\text{kin}} + \lambda I_J), \tag{6}$$

where w_{kin} controls coupling along the skeleton and $\lambda > 0$ stabilizes the inverse. Intuitively, Eq. (6) penalizes corrections that move a joint without coherently adapting its neighbors. Because the metric factorizes along the joint axis, R^{-1} can be applied efficiently by eigendecomposing the joint Laplacian once and reusing the result across frames and channels.

3.4. Pseudo-Observations for Sparse Controls

Sparse observations are difficult because a hard selector alone constrains too little of the motion early in sampling. This is particularly acute for long-gap inpainting. ProjFlow addresses this with time-varying pseudo-observations. Starting from the hard mask M_{obs}, we build a halo mask $M_{\text{halo}}(t)$ around observed frames by linear interpolation or extrapolation of the nearest anchors. The halo radius shrinks over time so that support is broad early in sampling and concentrates near the exact observations later.

Pseudo-observations should guide the sample but not dominate it. We therefore assign them adaptive uncertainty based on the curvature of the current clean estimate. For frame n, we compute a discrete acceleration score $s_n(\hat{x}_1)$ in the same kinematic metric and convert it into a frame trust value

$$\tau(t) = \tau_{\min} + (1-\tau_{\min})(1-t), \qquad \tilde{\pi}_n^{(t)} = \tau(t)\frac{c_0}{1 + \lambda_s(s_n/s_{\text{med}})^p}, \tag{7}$$

followed by clipping to $[\pi_{\min}, \pi_{\max}]$. The trust is then distributed across the halo joints using weights derived from the diagonal structure of R^{-1}. Finally, pseudo-observation variances are set as

$$\sigma_i^2(t) = r_i\left(\frac{1}{\pi_i^{(t)}} - 1\right), \tag{8}$$

where r_i is the corresponding diagonal entry of R^{-1}. Hard observations always keep zero variance. Figure 2 illustrates this mechanism.

3.5. Stochastic Recomposition and General Linear Operators

After projection, the sampler must return to the noisy manifold at the next time step. ProjFlow uses a FlowDPS-style stochastic recomposition [8]:

$$\tilde{x}_0 = \sqrt{1-\eta_t}\,\hat{x}_0 + \sqrt{\eta_t}\,\xi, \qquad x_{t+\Delta t} = \alpha_{t+\Delta t}x_1^\star + \sigma_{t+\Delta t}\tilde{x}_0, \tag{9}$$

with $\xi \sim \mathcal{N}(0, I)$ and $\eta_t = 1 - \sigma_{t+\Delta t}$ in our implementation. This step is crucial: without it, projection keeps the sample feasible but progressively pushes it off-manifold.

The inpainting sampler above corresponds to a selector operator A. For more general linear tasks, such as 2D-to-3D lifting, ProjFlow uses the same per-step logic but replaces the selector with a dense operator and precomputes a Cholesky factor of $AA^\top + \lambda I$. This yields a unified interface for trajectory control, loop closure, relative offsets, and orthographic 2D lifting in the provided codebase.

3.6. Implementation Details

The default sampler in the repository uses 100 flow steps. For the kinematics-aware metric, the implementation sets $w_{\text{kin}} = 10$ and $\lambda = 1$. For pseudo-observations, the halo radius shrinks linearly from $\ell_{\max} = 10$ frames to $\ell_{\min} = 3$ frames, and the trust schedule uses $\tau_{\min} = 0.1$, $c_0 = 3$, $\lambda_s = 1$, $p = 2$, $\pi_{\min} = 0.02$, and $\pi_{\max} = 1$. These choices are fixed at inference time and do not require task-specific tuning in the reported experiments.

For trajectory control and inpainting, the operator is a diagonal selector over the observed joint coordinates, so the projection system can reuse a shared observation-space factorization across channels. For general linear constraints, the implementation uses a dense operator with a precomputed normal-equation Cholesky factor. In the 2D lifting demo, this operator is an orthographic camera composed with joint selection and optional keyframe-0 anchoring. These details are important for reproducibility because they explain why ProjFlow supports exact control without any inner optimization loop.

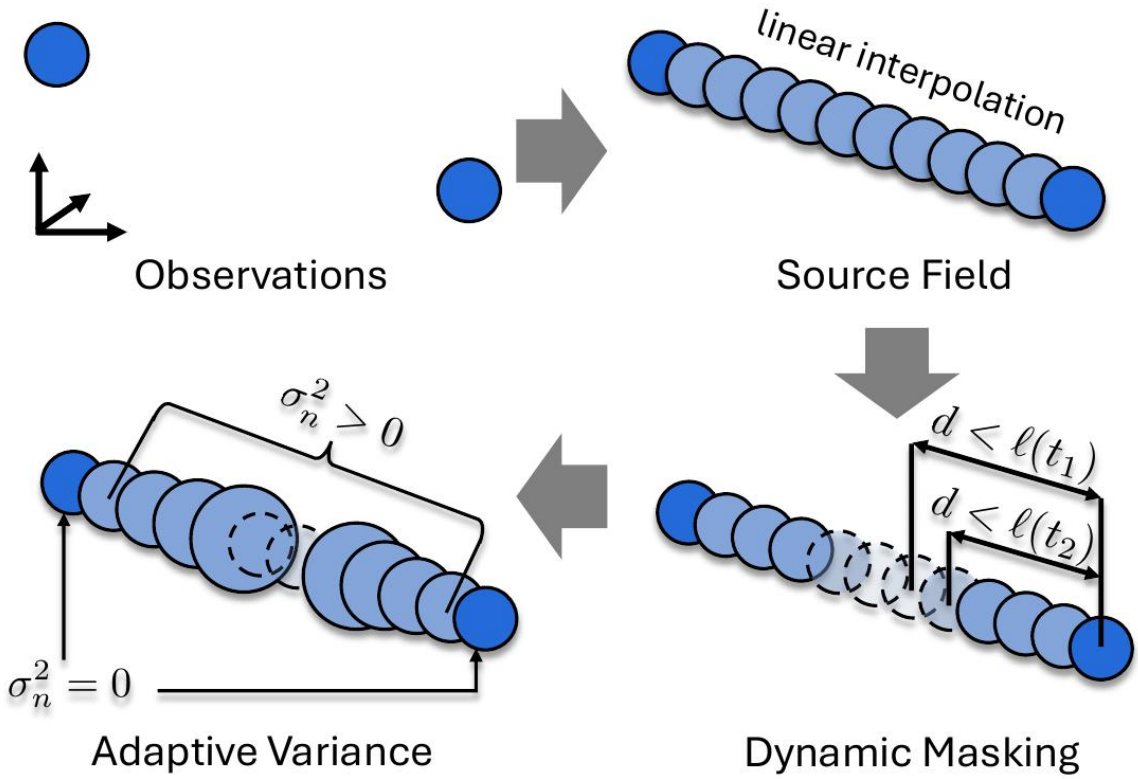


Figure 2. **Pseudo-observations for motion inpainting.** ProjFlow augments sparse hard observations with interpolated halo observations and time-varying uncertainty. The halo support is large at early steps and shrinks as sampling approaches the exact measurement set, giving the sampler broader temporal context before converging to the strict feasible set.

4. Experiments

4.1. Setup

All quantitative results are reported on HumanML3D [4] using the ACMDM-S-PS22 flow-matching backbone in the repository [11]. Motions use the standard 22-joint representation with up to 196 frames, matching the codebase and evaluation pipeline. We evaluate motion realism with FID, R-Precision, diversity, foot skating, and matching score, and evaluate control accuracy with trajectory/location errors or 2D/3D MPJPE depending on the task.

For trajectory control, the tables report two settings: pelvis-only control and control over arbitrary joints. We compare against zero-shot baselines and trained controllers reported in recent motion-control papers [3, 5, 6, 12, 18]. For 2D-to-3D lifting, we follow the protocol used in Sketch2Anim [21]; the table reports both the Average and Cross settings used by OmniControl [18]. The Average setting evaluates all constrained joints jointly, whereas Cross evaluates random combinations of constrained joints. We use the exact numbers provided by the experiment tables in the repository and do not infer any missing measurements beyond those tables.

4.2. Text-Conditioned Trajectory Control

Table 1 compares ProjFlow against trained and zero-shot baselines for pelvis-only control, all-joint control, and upper-body editing. The main result is that ProjFlow is the only method in the table that reaches 0.0000 trajectory, location, and average control error in both the pelvis-only and all-joint settings while remaining zero-shot. On pelvis control, ProjFlow achieves FID 0.107, substantially better than the other zero-shot baseline DNO at 0.151, while exactly satisfying the control signal. On all-joint control, ProjFlow attains FID 0.097 with exact control, again improving over DNO (0.147 FID) and avoiding the residual error of the trained ACMDM+CtrlNet controller (Avg. error 0.0197).

These results reveal a clear trade-off in prior work. Trained controllers can achieve slightly lower FID, for example ACMDM+CtrlNet reaches 0.070 on all-joint control, but they do not guarantee exact feasibility. MaskControl attains zero trajectory and location error yet still leaves a nonzero average error (0.0093 for pelvis and 0.0065 for all joints). ProjFlow closes this gap by delivering exact control

without task-specific training, with realism that stays in the same range as the best specialized models.

4.3. 2D-to-3D Lifting from Sketch-Like Constraints

Table 2 evaluates a harder inverse problem: generating 3D motions whose orthographic projections match 2D measurements. ProjFlow again enforces the linear measurement exactly, yielding 0.000 MPJPE-2D in both Average and Cross settings. In 3D, it achieves the best MPJPE-3D among the compared methods (0.042 and 0.037), outperforming Sketch2Anim's 0.048 and 0.046. It also improves realism over Sketch2Anim in FID, reaching 0.349 vs. 0.525 for Average and 0.168 vs. 0.577 for Cross.

The table also shows the limits of our zero-shot approach. Motion retrieval remains strongest on Cross FID (0.103) and foot skating (0.067), while Sketch2Anim retains the best text-motion matching and R-Precision. This is consistent with the task: ProjFlow enforces exact reprojection from sparse 2D evidence, which prioritizes geometric faithfulness over optimizing a learned sketch-specific alignment objective. Even so, the realism and text alignment remain competitive, and the exact 2D consistency is a strong practical advantage.

Comparing the Average and Cross settings also clarifies where projection sampling helps. ProjFlow preserves exact 2D feasibility in both regimes, but its realism improves markedly from 0.349 FID in Average to 0.168 in Cross. This suggests the sampler is not brittle to changing joint subsets and benefits when constraints still leave the prior enough freedom to resolve depth and pose naturally. The remaining gap in Avg. Err.-3D indicates the expected ambiguity of lifting from orthographic evidence: exact reprojection does not uniquely determine the unconstrained 3D structure.

4.4. Upper-Body Editing

Table 3 reports upper-body editing. Here ProjFlow is slightly behind the best trained controllers in realism and text alignment: it obtains FID 0.087, compared with 0.066 for MaskControl and 0.076 for ACMDM+CtrlNet. However, it remains competitive despite being zero-shot and preserves high diversity (10.571), which suggests that the projection mechanism does not collapse the motion manifold when the edit scope is localized to part of the body. This result is weaker than the trajectory-control results, which indicates that zero-shot projection is especially attractive when exact geometric constraints matter most.

5. Analysis

5.1. Ablation Study

Table 4 isolates the main design choices of ProjFlow. All variants keep exact control error at 0.0000, which confirms that the projection itself is sufficient for feasibility. The differences appear in realism. Replacing the kinematics-aware metric with a Euclidean metric degrades FID from 0.097 to 1.152, showing that geometry-aware correction is essential. Removing stochastic noise mixing is even more damaging, worsening FID to 3.429 and reducing diversity from 10.651 to 9.307; this supports our claim that recomposition is necessary to stay on-manifold after each projection. Finally, replacing pseudo-observations with plain masking increases FID to 0.880, indicating that sparse constraints benefit from the additional temporal support and adaptive uncertainty.

5.2. Why the Projection Works

The experiments support three complementary effects. First, exact control comes directly from the closed-form projection in clean-endpoint space, which is why all ProjFlow variants retain zero error. Second, the R-metric acts as a structural prior during correction: rather than moving only the observed joint, it distributes displacement across the kinematic tree, which is crucial for recovering natural whole-body responses. Third, stochastic recomposition decouples feasibility from realism. Projection enforces the measurements, while the recomposed noisy endpoint keeps the next iterate near the prior trajectory. This division of labor is visible in the ablations and mirrors the general intuition from inverse-problem solvers for diffusion and flow models [1, 8, 17].

The same perspective also explains when ProjFlow is less dominant. In upper-body editing and sketch-conditioned alignment, some baselines benefit from training on the specific control interface. A training-free sampler cannot exploit those paired signals directly; instead, it relies on the pretrained prior plus geometric consistency. The strong exact-control results nevertheless suggest that projection sampling is a compelling default when retraining is impractical or when hard feasibility is non-negotiable.

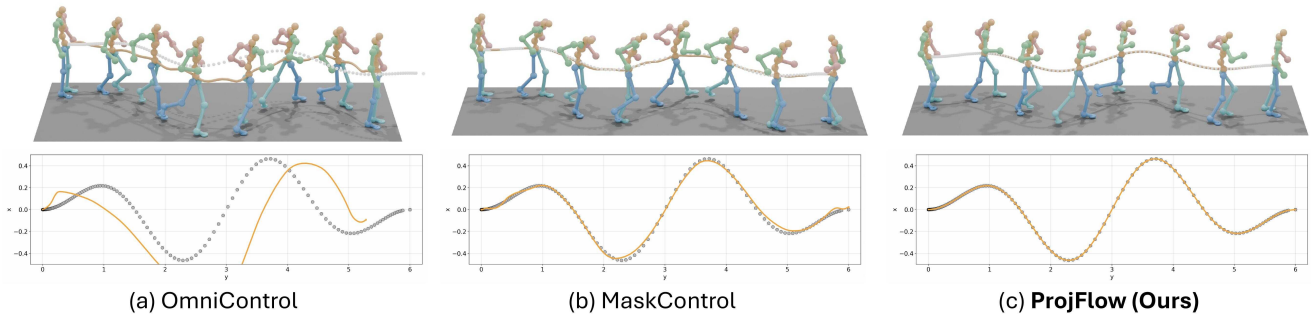
5.3. Practical Scope

Although our main quantitative study focuses on trajectory control and 2D lifting, the implementation exposes the same projection machinery for several additional linear tasks, including loop closure and relative joint offsets. We do not claim benchmark superiority on those tasks here because the repository does not provide comparable quantitative tables, but they are useful for understanding the method's scope. The key requirement is not a specific task family; it is the availability of a linear operator that maps a candidate motion to the observation space.

This perspective is important because it clarifies both the strength and the boundary of ProjFlow. When a control signal admits a reliable linearization, the method offers exact feasibility with essentially the same pretrained prior. When the constraint is intrinsically nonlinear or depends on

Table 1. **Quantitative text-conditioned motion generation with spatial control signals on HumanML3D[4].** In the first section, methods are trained and evaluated solely on pelvis controls. In the middle section, methods are trained on all joints and evaluated separately on each controlled joint. Only average results are reported for brevity. We include details in the supplementary material. ProjFlow is the only zero-shot method here with exact average control error in both sections, while keeping FID close to the strongest trained controllers. **bold** face / underline indicates the best/2nd results.

Controlling Joint	Methods	Zero-shot?	FID↓	R-Precision Top 3	Diversity→	Foot Skating Ratio.↓	Traj. err.↓	Loc. err.↓	Avg. err.↓
	GT	-	0.000	0.795	10.455	-	0.000	0.000	0.000
Pelvis	MDM [16]	✓	1.792	0.673	9.131	0.1019	0.4022	0.3076	0.5959
	PriorMDM [13]	✗	0.393	0.707	9.847	0.0897	0.3457	0.2132	0.4417
	GMD [5]	✓	0.238	0.763	10.011	0.1009	0.0931	0.0321	0.1439
	OmniControl [18]	✗	0.081	0.789	10.323	0.0547	0.0387	0.0096	0.0338
	MotionLCM V2+CtrlNet [3]	✗	3.978	0.738	9.249	0.0901	0.1080	0.0581	0.1386
	MaskControl [12]	✗	**0.066**	0.799	**10.474**	**0.0543**	**0.0000**	**0.0000**	0.0093
	ACMDM-S-PS22+CtrlNet [11]	✗	0.067	**0.805**	10.481	0.0591	0.0075	0.0010	0.0100
	ACMDM-S-PS22+DNO [6]	✓	0.151	0.802	–	0.0610	0.0027	0.0002	0.0089
	ACMDM-S-PS22+ProjFlow	✓	0.107	0.784	10.644	0.0629	**0.0000**	**0.0000**	**0.0000**
All Joints (Average)	OmniControl [18]	✗	0.126	0.792	10.276	0.0608	0.0617	0.0107	0.0404
	MotionLCM V2+CtrlNet [3]	✗	4.504	0.715	9.230	0.1119	0.2740	0.1315	0.2464
	MaskControl [12]	✗	0.095	0.795	10.159	**0.0545**	**0.0000**	**0.0000**	0.0065
	ACMDM-S-PS22+CtrlNet [11]	✗	**0.070**	**0.803**	**10.526**	0.0596	0.0117	0.0019	0.0197
	ACMDM-S-PS22+DNO [6]	✓	0.147	0.800	–	0.0600	0.0034	0.0003	0.0121
	ACMDM-S-PS22+ProjFlow	✓	0.097	0.779	10.651	0.0603	**0.0000**	**0.0000**	**0.0000**



(a) OmniControl (b) MaskControl (c) **ProjFlow (Ours)**

Figure 3. **Pelvis-trajectory control.** ProjFlow matches the controlled path exactly while preserving full-body adaptation under the prompt.

a learned semantic alignment objective, as in storyboard-specific sketch understanding, ProjFlow can still be applied only after that stage has been converted into linear measurements. In practice, this makes ProjFlow a strong back-end control mechanism rather than a replacement for every front-end conditioning model.

6. Conclusion

ProjFlow shows that exact spatial control for human motion can be achieved with a pretrained flow-matching prior and no task-specific retraining. By viewing motion control as a linear inverse problem, projecting each clean endpoint in a kinematics-aware metric, and recomposing the sample with stochastic noise, ProjFlow combines hard feasibility with competitive realism. On HumanML3D, it attains exact zero-shot trajectory control and exact 2D reprojection while remaining competitive with specialized trained controllers. These results suggest that projection-based flow sampling is a practical and general foundation for controllable motion generation.

Table 2. **Quantitative analysis of ProjFlow and three baseline models proposed in Sketch2Anim [21] on the HumanML3D [4].** Evaluation metrics on motion realism, control accuracy, and text-motion match are presented. Following OmniControl [18], we report both the average error of all joints (Average) and their random combination (Cross). ProjFlow uniquely attains zero 2D control error in both protocols and also improves MPJPE-3D over Sketch2Anim, although learned sketch-specific baselines still lead on some matching metrics. **bold** face / underline indicates the best/2nd results.

Condition	Method	Realism		Control Accuracy				Text-Motion Matching	
		FID ↓	Foot Skating ↓	MPJPE-2D ↓	MPJPE-3D ↓	Avg. Err.-2D ↓	Avg. Err.-3D ↓	Matching↓	R-precision (Top-3) ↑
Average	Motion Retrieval	0.690	**0.064**	0.057	0.076	0.290	0.410	4.060	0.640
	Lift-and-Control	0.979	0.089	0.054	0.071	0.261	0.340	3.297	0.752
	Direct 2D-to-Motion	2.553	0.112	0.040	0.055	0.193	0.275	3.723	0.687
	Sketch2Anim [21]	0.525	0.103	0.036	0.048	0.087	**0.134**	**3.077**	**0.802**
	ACMDM-S-PS22+ProjFlow	**0.349**	0.146	**0.000**	**0.042**	**0.000**	0.331	3.363	0.748
Cross	Motion Retrieval	**0.103**	**0.067**	0.055	0.073	0.307	0.423	3.405	0.724
	Lift-and-Control	0.738	0.101	0.051	0.067	0.209	0.283	3.135	0.778
	Direct 2D-to-Motion	2.310	0.123	0.040	0.056	0.189	0.266	3.606	0.709
	Sketch2Anim [21]	0.577	0.102	0.033	0.046	0.079	**0.132**	**3.042**	**0.796**
	ACMDM-S-PS22+ProjFlow	0.168	0.139	**0.000**	**0.037**	**0.000**	0.298	3.259	0.764

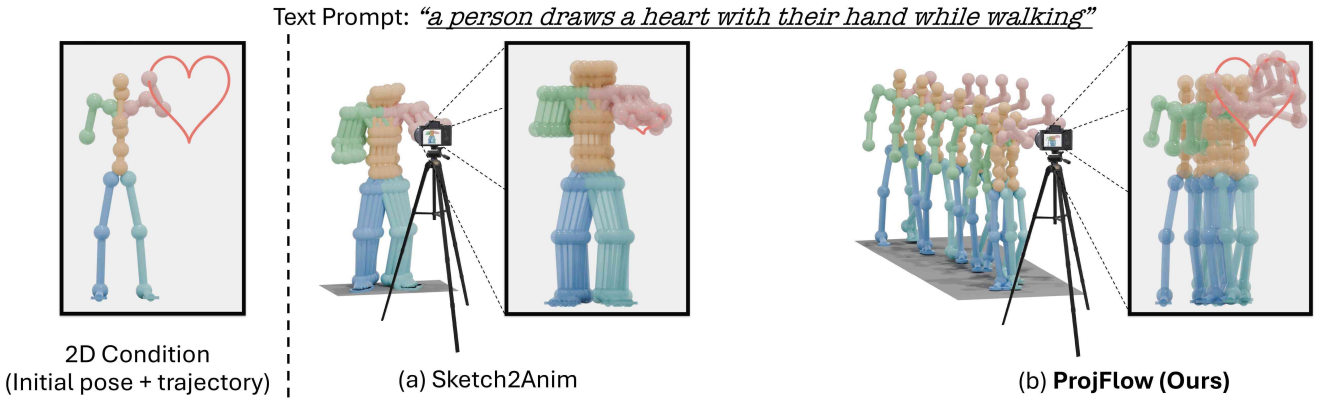


Figure 4. **2D-to-3D lifting.** Exact reprojection leaves depth ambiguity, but the pretrained motion prior resolves it into realistic 3D motion consistent with the prompt.

Table 3. **Upper-body editing on HumanML3D[4].** This setting is less favorable to zero-shot projection: ProjFlow remains competitive in FID and diversity, but the best trained controllers retain an advantage in text alignment. **bold** face / underline marks the best/2nd results.

	Methods	Zero-shot?	FID↓	R-Precision Top 1	R-Precision Top 2	R-Precision Top 3	Matching↓	Diversity→	–
Upper-Body Edit	MDM [16]	✓	1.918	0.359	0.556	0.654	4.793	9.210	–
	OmniControl [18]	✗	0.909	0.428	0.614	0.722	3.694	10.207	–
	MotionLCM V2+CtrlNet [3]	✗	3.922	0.404	0.592	0.692	5.610	9.309	–
	MaskControl [12]	✗	**0.066**	0.501	0.695	0.794	3.227	10.159	–
	ACMDM-S-PS22+CtrlNet [11]	✗	0.076	**0.532**	**0.719**	**0.820**	**3.098**	10.586	–
	ACMDM-S-PS22+ProjFlow	✓	0.087	0.501	0.690	0.787	3.319	**10.571**	–

Table 4. Ablation studies of ProjFlow. All variants preserve zero control error, so the realism gap isolates the contribution of each design choice. The sharp drop without stochastic recomposition shows that feasibility alone is insufficient for realistic motion.

Variant	FID↓	R-Prec.	Div.→	Foot↓	Traj.↓	Loc.↓	Avg.↓
ProjFlow (Full)	0.097	0.779	10.651	0.0603	0.0000	0.0000	0.0000
Euclid. (R=I)	1.152	0.740	10.107	0.0595	0.0000	0.0000	0.0000
No noise (η_t=0)	3.429	0.707	9.307	0.0863	0.0000	0.0000	0.0000
Plain masking	0.880	0.748	10.187	0.0632	0.0000	0.0000	0.0000

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